Data Card & Datasheet for NYC Spatial Mobility & Socioeconomic Equity Datasets

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Project: Geographic and Socioeconomic Bias in Spatial Mobility Data
Course: Foundations of Big Data & Responsible AI Engineering (Topic 20)
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Standard Compliance: Gebru et al. (*Datasheets for Datasets*) & Mitchell et al. (*Model Cards for Model Reporting*)

1. Dataset Motivation

- **For what purpose was the dataset created?**
  To empirically measure, benchmark, and mitigate geographic and socioeconomic bias in urban on-demand mobility and delivery services across New York City's 5 boroughs and 71 TLC Taxi Zones.
- **Who created the dataset?**
  Compiled and synthesized by the research team using publicly available open government and spatial repositories:
  - New York City Taxi and Limousine Commission (NYC TLC Open Data)
  - United States Census Bureau (American Community Survey - ACS 5-Year Estimates)
  - OpenStreetMap (OSM) via Geofabrik spatial extracts
- **Who funded the creation of the dataset?**
  Academic project conducted at PES University under the Foundations of Big Data & Responsible AI Engineering curriculum.

2. Composition & Data Schema

- **What do the instances that comprise the dataset represent?**
  1. **Trip Records (`nyc_trips_scaled.csv` - 250,000 rows):** Individual on-demand dispatch logs including `trip_id`, `PULocationID`, `DOLocationID`, `pickup_hour`, `trip_distance_miles`, `wait_time_minutes`, `trip_duration_minutes`, `fare_amount`, `tip_amount`, `total_amount`, and `trip_status`.
  2. **Spatial Zone Lookup (`taxi_zone_lookup.csv` - 71 zones):** TLC Zone ID, Borough, Zone Name, Service Zone, Centroid Latitude, and Centroid Longitude.
  3. **Census Demographics (`nyc_census_acs_zones.csv` - 71 zones):** Median household income, poverty rate, minority population percentage, zero-car household percentage, total population, and transit dependence index.
  4. **OpenStreetMap Infrastructure (`nyc_osm_infrastructure.csv` - 71 zones):** Road network length (km), subway station counts, commercial POI counts, intersection density per sq km, and walkability scores.
- **Does the dataset contain all possible instances or is it a sample?**
  Represents a scaled, statistically balanced 250,000-trip cohort across peak and off-peak travel periods reflecting empirically validated NYC TLC trip distribution and spatial queuing behaviors.
- **Are there sensitive attributes or personal data included?**
  No. In strict adherence to Responsible AI and project guidelines:
  - **Zero Personally Identifiable Information (PII):** No passenger names, driver IDs, credit card details, IP addresses, or exact GPS trajectories.
  - Aggregated strictly at the spatial TLC Taxi Zone level (centroid coordinates).

- Further protected via ϵ-**Differential Privacy (Laplace Mechanism)** for public query release.

3. Collection & Synthesis Methodology

- **How was the data acquired or generated?**
  - Real TLC Taxi Zone lookup tables and spatial centroid definitions were mapped to 2020 US Census PUMA and ACS economic tables.
  - OpenStreetMap Overpass API features were extracted for transportation network metrics (transit stops, road density).
  - Spatial queuing and market equilibrium algorithms synthesized trip dispatch logs calibrated against empirical NYC TLC trip frequency, driver concentration in commercial hubs, and outer-borough dispatch friction.
- **What preprocessing was performed?**
  - Data cleaning to eliminate impossible spatial coordinates and invalid durations.
  - Feature normalization for per-capita metrics (Trips$/1,000$ residents).
  - Construction of the unified spatial equity feature store (`zone_equity_features.csv`).

4. Uses & Intended Applications

- **What are the intended uses?**
  - Academic research on algorithmic fairness in spatial queuing systems.
  - Benchmarking Big Data distributed pipelines (PySpark, Lakehouse, Spark Structured Streaming).
  - Policy simulation for transportation regulators (e.g., NYC TLC, transit authorities) seeking to audit algorithmic dispatch systems for compliance with civil rights statutes (EEOC 80% Rule, Title VI, NYC Local Law 144).
- **What are the out-of-scope or prohibited uses?**
  - **Do NOT** use to target individual driver earnings, blacklist low-income neighborhoods, or justify discriminatory redlining.
  - **Do NOT** use without differential privacy noise addition when operating on small demographic slices.

5. Distribution & Licensing

- **How is the dataset distributed?**
  Hosted directly inside the reproducibility repository under the `data/` directory.
- **What is the license?**
  - NYC Open Data Terms of Use (Public Domain).
  - OpenStreetMap Open Database License (ODbL).
  - Code & Derivatives: MIT License.

6. Maintenance & Governance

- **Who maintains the dataset?**
  Namith Kumar C (`PES1UG24AM171`) & Jayanth K (`PES1UG24AM123`).
- **Governance & Errata:**
  Any corrections or schema evolutions are tracked in git commits with OpenLineage and Lakehouse versioning principles.